

week6exercise

February 8, 2026

1 For PF Section

2 Data Visualization Week-6 Lab exercise

- BTech Computer Science Stream , Februry 2026
- Week 6 - Data Visualization: context, effective visuals and storytelling
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- Date: 12/02/2026
- Data File: mtcars
- Metadata: <https://rpubs.com/neros/61800>

This notebook uses a well known dataset called mtcars. Always read the metadata thoroughly to understand the dataset. The focus of this notebook is to perform a. Data exploration using Descriptive Statistics b. Data Cleaning and preparation c. Basic Data Analysis & Visualization to understand relationships among features of cars

```
[1]: import pandas as pd
```

```
[7]: dat_mtcars=pd.read_csv("mtcars.csv", index_col=0)
```

```
[13]: dat_mtcars
      #dat_mtcars.head()
```

```
[13]:
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	\
model											
Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	
Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	
Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	
Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	
Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	


```
carb
```

model	
Mazda RX4	4
Mazda RX4 Wag	4
Datsun 710	1
Hornet 4 Drive	1

```
[9]: # Descriptive Statistics
# info() provides a concise summary of the data frame in terms of the columns,
# datatype and memory usage
dat_mtcars.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 32 entries, Mazda RX4 to Volvo 142E
Data columns (total 11 columns):
#   Column  Non-Null Count  Dtype
---  -
0   mpg     32 non-null      float64
1   cyl     32 non-null      int64
2   disp    32 non-null      float64
3   hp      32 non-null      int64
4   drat    32 non-null      float64
5   wt      32 non-null      float64
6   qsec    32 non-null      float64
7   vs      32 non-null      int64
8   am      32 non-null      int64
9   gear    32 non-null      int64
10  carb    32 non-null      int64
dtypes: float64(5), int64(6)
memory usage: 3.0+ KB
```

```
[10]: # Descriptive Statistics
# describe() is used to understand the distribution of the data , 5 number
# summary , mean and standard deviation
dat_mtcars.describe()
```

```
[10]:
```

	mpg	cyl	disp	hp	drat	wt \
count	32.000000	32.000000	32.000000	32.000000	32.000000	32.000000
mean	20.090625	6.187500	230.721875	146.687500	3.596563	3.217250
std	6.026948	1.785922	123.938694	68.562868	0.534679	0.978457
min	10.400000	4.000000	71.100000	52.000000	2.760000	1.513000
25%	15.425000	4.000000	120.825000	96.500000	3.080000	2.581250
50%	19.200000	6.000000	196.300000	123.000000	3.695000	3.325000
75%	22.800000	8.000000	326.000000	180.000000	3.920000	3.610000
max	33.900000	8.000000	472.000000	335.000000	4.930000	5.424000

	qsec	vs	am	gear	carb
count	32.000000	32.000000	32.000000	32.000000	32.0000
mean	17.848750	0.437500	0.406250	3.687500	2.8125
std	1.786943	0.504016	0.498991	0.737804	1.6152
min	14.500000	0.000000	0.000000	3.000000	1.0000
25%	16.892500	0.000000	0.000000	3.000000	2.0000

50%	17.710000	0.000000	0.000000	4.000000	2.0000
75%	18.900000	1.000000	1.000000	4.000000	4.0000
max	22.900000	1.000000	1.000000	5.000000	8.0000

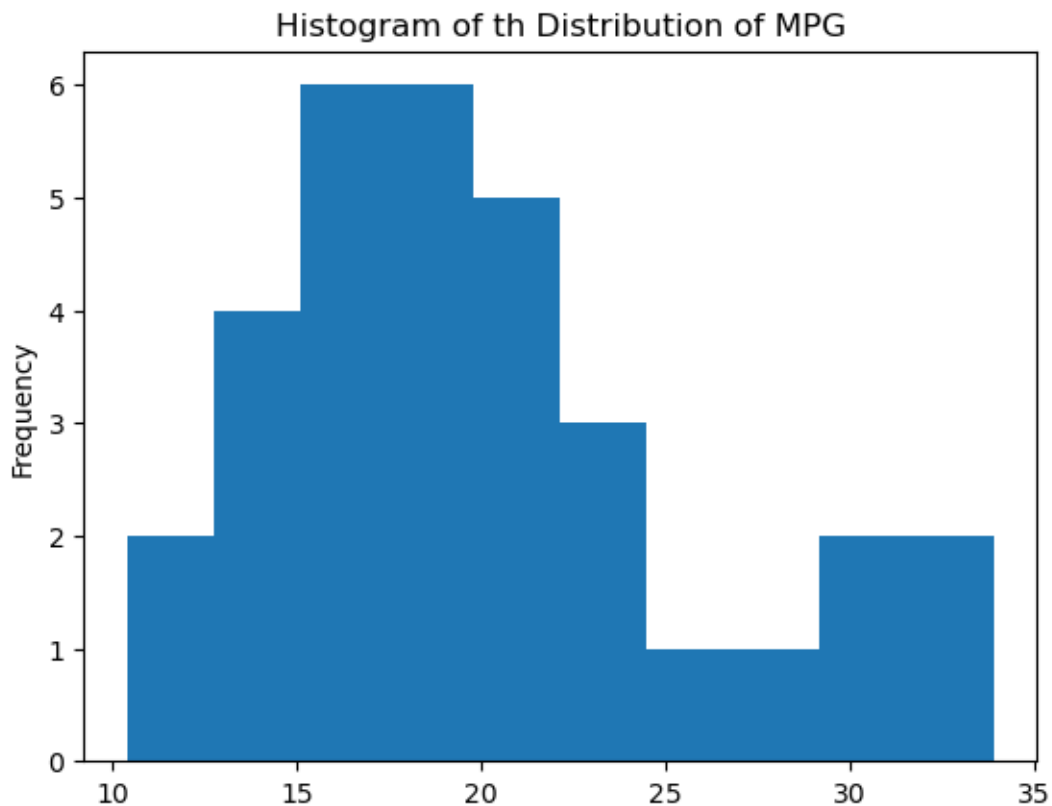
```
[12]: # Descriptive Statistics
      # Use shape to show number of rows and columns
      dat_mtcars.shape
```

```
[12]: (32, 11)
```

```
[ ]: # Data Cleaning - look for missing values, replaced with mean of the column,
      ↪ look for noisy values, replace with mean
```

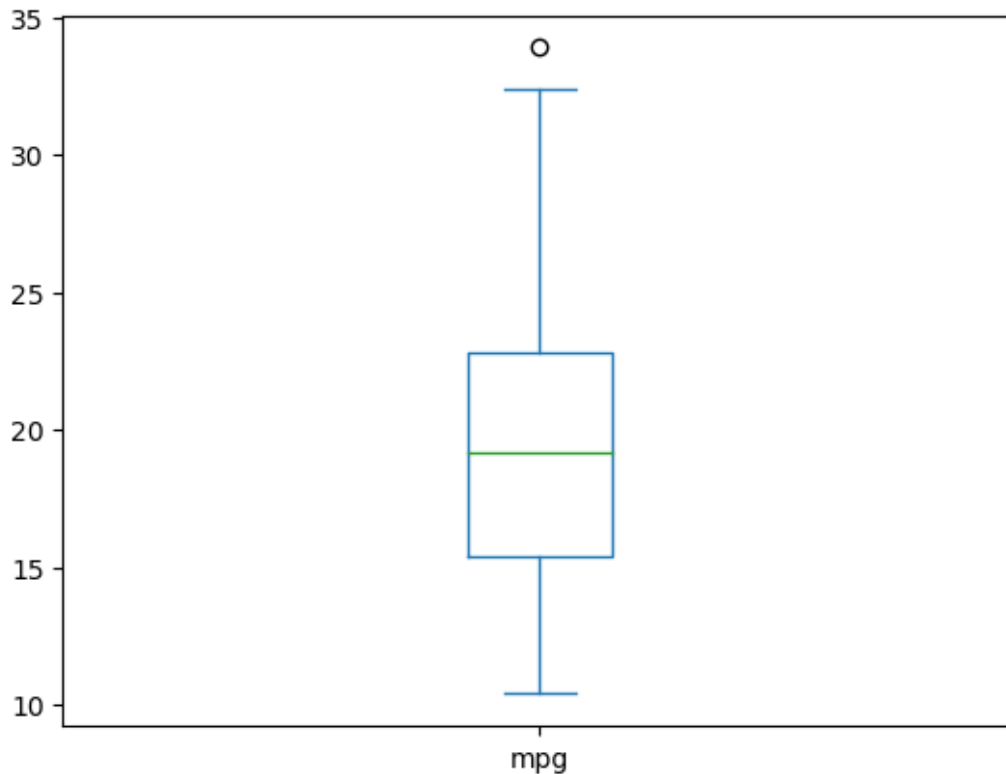
```
[17]: # Data Analysis & Visualization
      # 1. Explore the distribution of Fuel efficiency of the cars
      # Remember to label both axis and put a title to the plot
      dat_mtcars["mpg"].plot(kind='hist', bins=10, title="Histogram of th_
      ↪ Distribution of MPG")
```

```
[17]: <AxesSubplot:title={'center':'Histogram of th Distribution of MPG'},
      ylabel='Frequency'>
```



```
[18]: # Data Analysis & Visualization
# 1. Explore the distribution of Fuel efficiency of the cars
dat_mtcars["mpg"].plot(kind='box')
```

```
[18]: <AxesSubplot:>
```



```
[23]: # Data Analysis & Visualization
# 2. Find the cars with the best and worst fuel efficiency
max_mpg=dat_mtcars["mpg"].max()
dat_mtcars[dat_mtcars["mpg"]==max_mpg]
```

```
[23]:
```

	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
model											
Toyota Corolla	33.9	4	71.1	65	4.22	1.835	19.9	1	1	4	1

```
[26]: # Data Analysis & Visualization
# Features like am, gears are numeric but binary features. They are used to
→actual categorise cars into types.
# 3. How many cars are manual vs automatic?
dat_mtcars["am"].nunique()
```

```
[26]: 2
```

```
[28]: dat_mtcars["am"].value_counts()
```

```
[28]: 0    19  
      1    13  
      Name: am, dtype: int64
```

```
[ ]: # Data Analysis & Visualization  
      # 4. Find the car with the worst horsepower  
      # 5. Find 5 number summary and draw the boxplot of displacement.  
      # 6. Which is the heaviest car? How many gears does it have?  
      # 7. Which is the car with the best qsec?
```

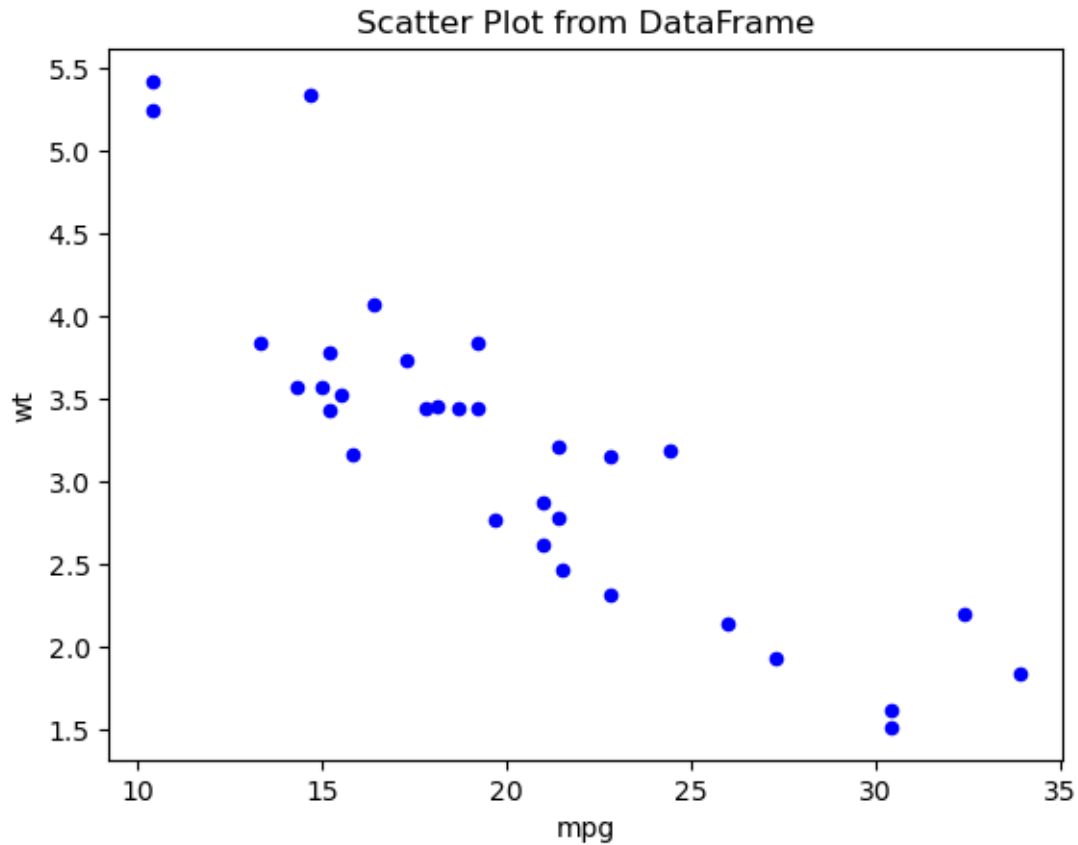
```
[38]: # Data Analysis & Visualization  
      # 8. What is average MPG for manual vs. automatic cars?  
      dat_man= dat_mtcars[dat_mtcars["am"]==1]["mpg"] # Manual cars are marked as 1  
      print("Average MPG of manual cars:",dat_man.mean())  
      dat_auto= dat_mtcars[dat_mtcars["am"]==0]["mpg"]  
      print("Average MPG of automatic cars:",dat_auto.mean())
```

```
Average MPG of manual cars: 24.39230769230769  
Average MPG of automatic cars: 17.147368421052633
```

```
[39]: # Data Analysis & Visualization  
      # 9. Draw Side by Side box plot to understand the difference in fuel efficiency  
      ↪ of Manual vs Automatic cars. Analyze and write about fuel efficiency in  
      ↪ each group (manual vs. automatic).
```

```
[57]: # Data Analysis & Visualization  
      # 10. What is the relationship between the weight of the car and MPG?  
      # Create scatter plot  
      # Remember to label both axis and put a title to the plot, write the units too  
      dat_x= dat_mtcars[["mpg","wt"]]  
      dat_x= dat_x.reset_index(drop=True)  
      dat_x.plot(kind='scatter', x='mpg', y='wt', color='blue', title='Scatter Plot  
      ↪ from DataFrame')
```

```
[57]: <AxesSubplot:title={'center': 'Scatter Plot from DataFrame'}, xlabel='mpg',  
      ylabel='wt'>
```



- []: *# Data Analysis & Visualization*
 # 11. Categorize the cars based on the number of gears in the cars. How many ↵
 ↪ cars are there in each type?
- []: *# Data Analysis & Visualization*
 # 12. What is the relationship between fuel efficiency and the number of gears ↵
 ↪ in the car?
- []: *# Data Analysis & Visualization*
 # 13. Explain the relationship between horse power and number of cylinders in ↵
 ↪ the car.
- []: *# Data Analysis & Visualization*
 # 14. Explain the relationship between displacement and gross horse power.
- []: *# Data Analysis & Visualization*
 # 15. Which car would I pick if I am looking for high speed as well as good ↵
 ↪ fuel efficiency?

2.0.1 Lab Questions (MTCARS Dataset)

1. Load the MTCARS dataset and display its basic structure (rows, columns, data types).
2. Compute summary statistics (mean, median, min, max) for:
 - Miles per gallon (mpg)
 - Horsepower (hp)
 - Weight (wt)
3. Identify the relationship between engine cylinders (cyl) and fuel efficiency (mpg) using an appropriate visualization.
4. Create a scatter plot between horsepower (hp) and mpg.
 - Add labels and a meaningful title.
5. Compare manual vs automatic transmission cars in terms of mpg using a box plot.
6. Identify which car model gives the best balance of power and fuel efficiency. Justify using a visualization.
7. Write a short data story (3–4 lines) summarizing insights derived from the plots.

2.0.2 Lab Questions (CEREALS Dataset)

1. Load the CEREALS dataset and inspect missing values.
2. Calculate average calories, sugar, and fiber content.
3. Create a bar chart showing top 5 cereals with highest sugar content.
4. Visualize the relationship between calories and rating.
5. Compare cold vs hot cereals based on calorie distribution.
6. Identify healthier cereals using at least two nutritional parameters.
7. Summarize findings using one key visualization and explanation.

[]: